Multimodal Emotion Detection from Image and Speech Using Advanced Deep Learning

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ABSTRACT

Affective computing or emotion recognition is the automatic perception and interpretation of human affective conditions based on inputs like speech and facial expressions. The paper will present a multimodal emotion recognition system that will combine Swin Transformers as the facial expression recognition (FER) system, HuBERT as the speech emotion recognition (SER) system, and cross-modal attention and graph neural networks (GNN) to fuse the two systems. The system is tested using three benchmark datasets, namely, RAF-DB, RAVDESS, and FER2013, and is consistently improving baselines by 46% on unimodal systems. The paper discusses strength, imbalanced classes, equity, interpretability, and real-time executability on edge gadgets. The main issues such as cross-dataset generalization, demographic bias, and privacy are addressed as well as ethical aspects and a roadmap of future research are presented.

Keywords- emotion recognition; multimodal learning; Swin Transformer; HuBERT; speech emotion recognition; facial expression recognition; deep learning; cross-modal attention; graph neural network; affective computing.

I. INTRODUCTION

Emotion recognition also known as affective computing is the automated process of recognizing and interpreting human affective states among various modalities such as facial expressions, speech patterns, body movements and physiological cues. As human- computer interaction becomes increasingly more technologically integrated, machine system abilities to recognize and respond to human emotions have evolved into a wanted feature to a needed feature.

The default architectural structures in visual emotion recognition have become Convolutional Neural Networks (CNNs). Conventional frameworks like ResNet, VGG, and EfficientNet are potent backbones with respect to extracting features of face images. Long Short-term Memory (LSTM) networks are used to model the temporal dependencies of speech signals to identify emotions. HuBERT and Wav2Vec are self-supervised models that do not require labeled audio inputs and instead learn on large volumes of unlabeled audio data and offer rich features that are more effective than hand-designed features. Vision Transformers (ViTs) and Swin Transformers use self-attention to discover the long-range interactions and holistic attributes of the facial expression.

The problems are still there, despite these developments. The bias of datasets causes models to be good at predicting common affective states and bad at predicting rare affective states. The cross-domain generalization is low when models are trained on one data set and tested on another with different recording conditions. Demographic and cultural prejudices restrain the generalization to non-Western populations. An issue of privacy and ethics is the constant monitoring of emotions and its possible abuse to monitor or discriminate.

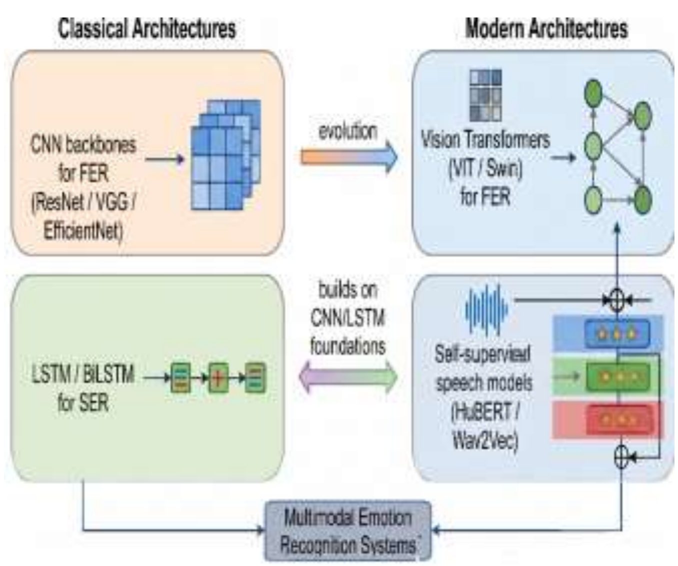


Fig. 1. Evolution of Emotion Recognition: Classical vs. Modern Deep Learning Architectures.

Scopes and Objectives of this Review

- Facial expression and speech emotion recognition state of the art architecture.
- Comparison of single-modality and multimodal fusion

- Large publicly accessible datasets and their properties.
- The suitable evaluation methodologies and performance measures regarding the imbalanced classification of emotions.
- Effects and limitations of practical implementation.
- Social Consequences and Ethics.

Figure 1 describe the **e**volution of emotion recognition of the classical and modern deep learning architectures. The emotion recognition field has made significant advances since its manual feature engineering systems to deep learning models that combine the use of several modalities. Vision Transformers and self-supervised models have replaced the fundamental role of CNNs to extract spatial features out of face images and LSTM to model temporal dynamics of speech signals, although not eliminated. Modern state-of-the-art systems take advantage of the complementary capabilities of CNNs to extract local features, LSTMs to model time, and Transformers to understand the global context.

II. LITERATURE REVIEW

The recent progress made on facial emotion recognition has demonstrated great improvements on performance through the integration of attention mechanisms with the conventional CNN models. Hu et al. showed that Squeeze-and-Excitation (SE) blocks and Convolutional Block Attention Modules (CBAM) enhance informative facial regions and reduce irrelevant background components that boost recognition performance, especially in the case of occlusions, varying illumination and cluttered backgrounds.

Liu et al. [3] introduced the Swin Transformer that uses shifted-window attention to create a hierarchical structure that balances local image detailing and global awareness. Swin architecture demonstrated state-of-the-art results on the AffectNet large-scale dataset of more than one million images, and it outperforms the traditional CNN architectures on in-the-wild datasets and is more sensitive to subtle facial micro-expressions and emotion fusions. HuBERT is trained with thousands of hours of unlabeled Librispeech audio without the need to manually engineer features to obtain rich acoustic representations. On emotion recognition tasks when fine-tuned, HuBERT significantly outperforms systems using standard handcrafted acoustic features and systems based on LSTMs, captures fine prosodic variations unnoticed with traditional feature extraction, and cross-adapts effectively across speakers and across languages.

The LSTM and BiLSTM networks are still of great significance towards discovering temporal relationships in speech signals. Attention systems in speech emotion recognition identify and draw attention to emotionally salient stretches of time like intensity peaks, pauses or rapid variations in pitch. Wang used dynamic fusion whereby the model gives importance to audio or visual data depending on how reliable they are in a particular context. Tsai disclosed that transformer networks have the capacity to depict connections among video frames, speech signals and text transcriptions. Lane found that there were severe cultural biases in widely used emotion datasets that led to systematic misclassification of non-Western populations.

Wang uses dynamic fusion, whereby the model is conditioned to give importance to audio as well as visual data depending on how reliable they are under a particular context in multimodal fusion research. In case of high audio and low facial expression the system itself enhances the reliance on visual data and the reverse. It was experimented with the IEMOCAP data and proved to be far better than fixed fusion policies. The other innovation that has since ensued has been the consideration of audio and visual pieces as nodes in a graph where temporal and cross-modal dependencies are coded as edges. Graph neural networks consist of such connections in which information flows and reflect multifaceted non-local interactions between modalities. This model is particularly effective in the forecasting of the valence-arousal on the MSP-IMPROV data.

As the literature shows, Vision Transformers and self-supervised speech models are far more successful than their traditional counterparts, such as ResNet, VGG, LSTM, and BiLSTM; nevertheless, their traditional counterparts still form a crucial part of which systems are being developed nowadays. The stepwise incorporation of attention modules into the convolutional networks, the application of the LSTM layer to introduce temporal discourse in the auditory sensory modalities and the application of the multimodal fusion paradigm, still give better performance. There are still significant loopholes, however, in cross dataset generalization, demographic fairness and real-world implementation. It is based on this that the field has made a bold move towards end-to-end deep-learning systems that are more focused on interpretability, fairness and computational efficiency.

III. PROBLEM FORMULATION

The biggest challenge of emotion recognition is the uncertainty and subjectivity of human expression. Emotions are not heterogeneous across people, cultures, and circumstances but complex psychological states that are not discrete phenomena. Research gaps have been identified as: (1) Robustness and Generalization - high accuracy on benchmark datasets plummets in new settings; (2) Temporal Dynamics and Context - current systems are to a large extent operating on static snapshots, instead of modeling how emotion evolves over time; (3) Privacy sensitive emotion recognition current systems are required to support user privacy and still be useful.

The problem statement involves creating a multimodal emotion recognition model that: can be used in varying lighting, pose, background noise, and demographics; can be trained to balance the class imbalance using weighted loss functions and data augmentation; can be trained to combine facial and speech performance with adaptive fusion; can be trained to run with a latency of less than 200ms and a model size of less than 100MB.

The primary objectives are to develop a state of the art multimodal emotion recognition framework based on the most current architectures such as Vision Transformers, self-supervised speech models, cross-modal transformers to provide high-performances. The system needs to face the real-world robustness issues and assesses its performance on a variety of datasets that are different in acquisition conditions, implements domain adaptation methods to cross-dataset generalization, and performs on realistic conditions with changing lighting, pose, and background noise.

It needs to correct class imbalance by weighted loss functions that penalise misclassification of rare emotions, data augmentation policies that are specific to affective data, and suitable evaluation metrics like Unweighted Average Recall (UAR), F1 -score that take into account class imbalance It needs to be more practical to deploy, through real-time performance with less than 200ms latency, knowledge distillation and quantisation to reduce model size, and be able to process on device to protect privacy. It must be explainable and fair, making sure that attention-based visualisations of decision-critical features, between-demographic-group bias audits, and loss functions and debiasing procedures based on fairness are considered.

The problem statement shows that there are grave challenges in emotion recognition on technical aspects such as generalization and multimodal synchronization, issue of fairness of discrimination based on demographics, privacy of continuous surveillance as well as the issues of practicality in actual practice. The identified problem statement and research gaps provide clear goals to be developed around which a system can be developed that is not only capable of achieving state-of-the art performance but can also be capable of addressing much broader overarching issues of robustness, fairness, explain ability, and practical deploy ability.

IV. PROPOSED WORK

A. Justification for Proposed Components

The rationale behind the proposed multimodal emotion recognition system is supported by the findings of literature review. Using shifted-window attention-based swin transformers, local detail extraction of facial micro-expressions and global context of overall facial structure, achieves greater performance on AffectNet and in- the-wild tasks. HuBERT fine-tuned to SER achieves invariably better results than CNN+LSTM baselines. Cross-modal attention fusion is a dynamically fusing audio and visual modalities in response to input quality - favouring visual in cases of noisy audio, and audio in cases of occluded faces. Fairness is directly measured between demographic subgroups using adversarial debiasing.

B. System Pipeline

Step 1 - Preprocessing: Face recognition with RetinaFace (confidence threshold 0.8) and face alignment with 5-point landmarks to 224x224 canonical orientation, histogram equalisation or CLAHE illumination normalisation, audio-video time warping.

Step 2 - Facial Feature Extraction: The 768-dimensional projections of the frames are divided into 16x16 patches and the output of a Swin Transformer backbone with four stages using shifted-window attention (7x7 window size), a spatial attention unit with CBAM channel and spatial importance weights, and temporal position encoding to generate feature sequences of shape (T, 768).

Step 3 - Speech Feature Extraction: Mel-spectrograms are calculated using 128 mel-bands, 2048 FFT-size, and 512 hop-size (32ms resolution). The HuBERT encoding is a 12-layer transformer pre- trained on unlabeled sound, fine-tuned on emotion-labeled data. Temporal context is represented by a two-layer BiLSTM (256 hidden units per direction, 0.3 dropout). Temporal attention emphasizes emotionally salient audio frames, and the output sequences are of shape (T spec, 512).

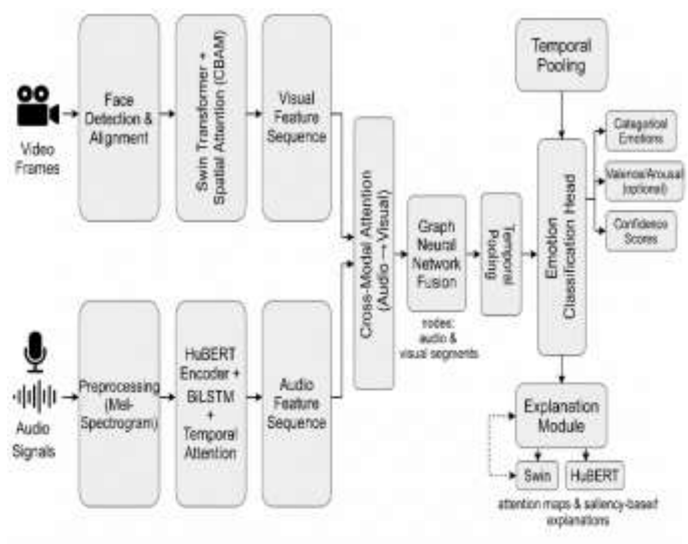


Fig. 2. Proposed Multimodal Emotion Recognition Architecture

Step 4 - Cross-Modal Attention Fusion: Multi-head cross- attention (8 heads) is performed in both directions of visual attending to audio and audio attending to visual, learning to align in different ways at the same time.

Step 5 - GNN Fusion: Audio and visual temporal segments are modeled as nodes; an edge between nodes is learned through MLPs, which encode inter-modality interactions; message passing through 3 layers aggregates to form a 768-dimensional feature.

Step 6 Classification: Use fully connected layers (7685122565) with ReLU and dropout (0.5), then use softmax, to get emotion probabilities and confidence scores.

Step 7 - Explanation Generation: The weight of spatial attention of the Swin Transformer is upsampled to frame resolution in the form of heatmap overlays. Temporal saliency is computed via gradient computation identifying impactful audio frames

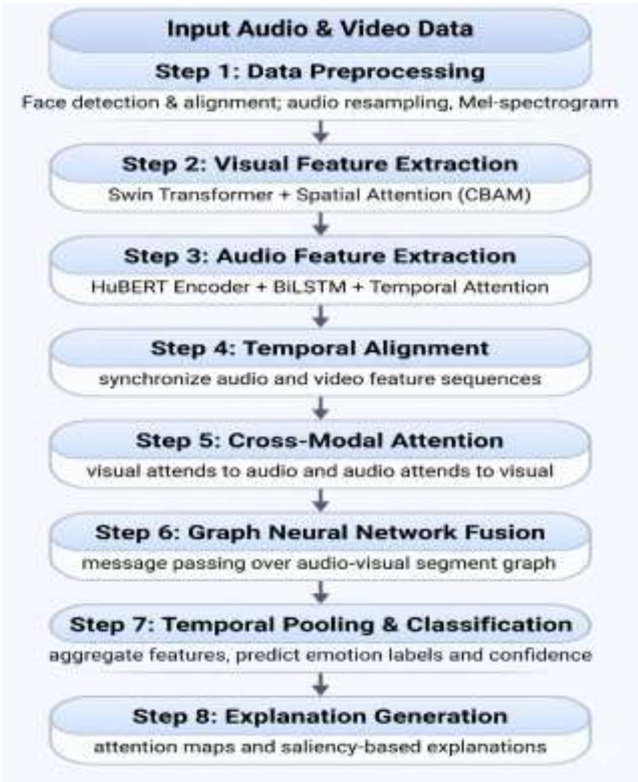


Fig. 3. Algorithm Flowchart: of Multimodal Emotion Recognition Process

Figure 2 shows the proposed multimodal emotion recognition architecture and **figure 3** describe the dataflow of the multimodal emotion recognition system

This paper lays a substantive multimodal emotion-recognition system by drawing in some of the most recent technologies, such as Swin Transformers, HuBERT, cross-modal attention, and graph neural nets. Our architecture is selected according to the recent researches to address the known issues in terms of robustness, generalization, fairness, and explain ability.

It is because of the comprehensive algorithm, the step-by-step plan that we are able to start with raw inputs and finally predict emotions, which have been cleared and assessed with confidence scores. All these recall the limitations of ethics and a real-life deployment and hence the system is able to score above the benchmarks and real- life conditions.

V. SYSTEM DESIGN AND ARCHITECTURE

It is a modular, multimodal emotion recognition pipeline, which processes auditory and visual data, and produces emotion categories, confidence scores, and interpretable explanations. At the macro level, the system decodes raw multimodal data to emotion predictions through deep learning, and gives attention visualization and saliency-centric explanations to enhance transparency and comprehension.

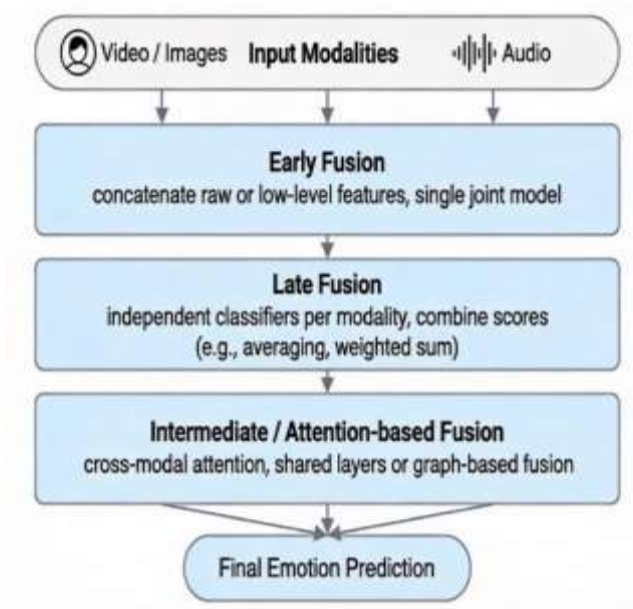


Fig. 4. Multimodal Fusion Strategies: Early, Late, and Attention-based Fusion**.**

The system is created to be a modular structure where the primary controller is the Emotion Recognition System which includes a Visual Encoder, Audio Encoder, Fusion Module and Classifier. It facilitates mixed deployment: either inference can be done on the edge with quantized models in privacy sensitive settings, or on a cloud server through REST APIs with more computing capabilities. **Figure 4** describe the multimodal fusion strategies while **figure 5** describe the deployment architecture for multimodal emotion recognition.

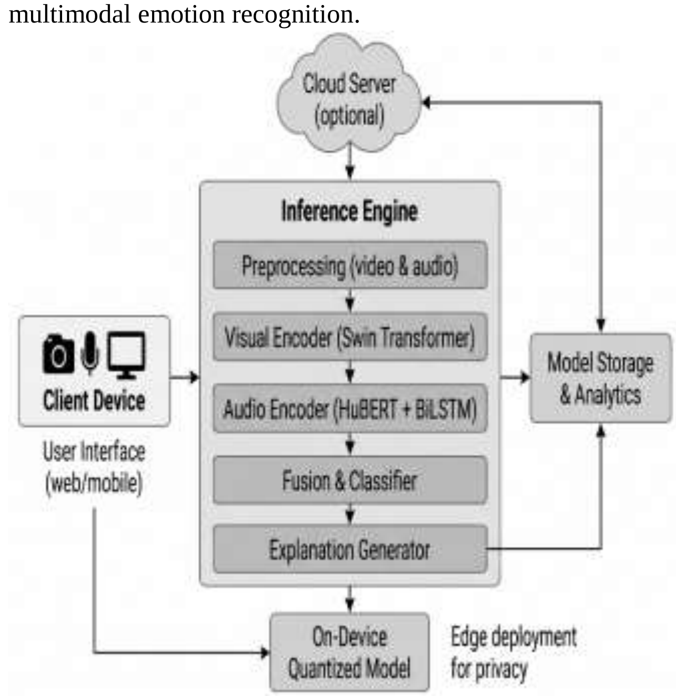


Fig. 5. Deployment Architecture for Multimodal Emotion Recognition

Multimodal pre-processing activates the data flow including parallel feature extraction, temporal alignment, fusion of cross-modal attention, GNN fusion, classification, and explanation generation

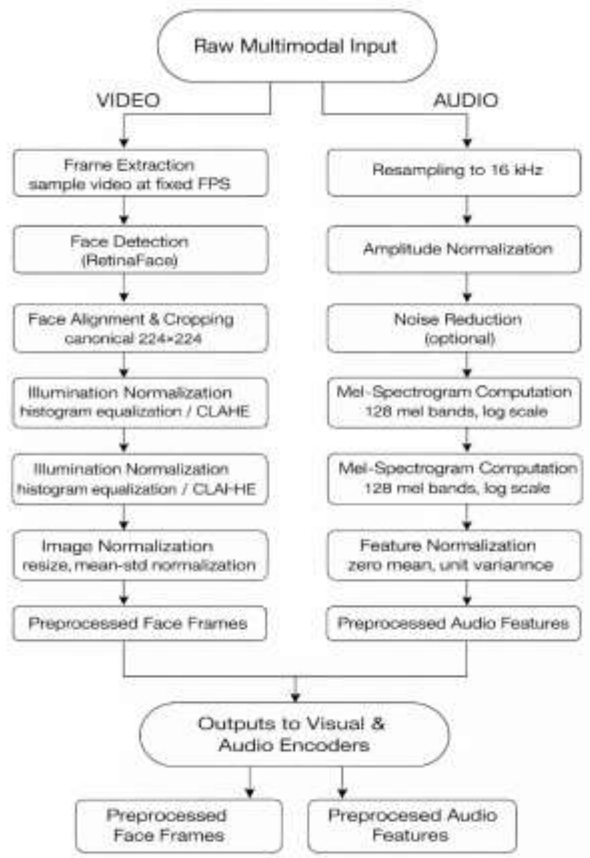


Fig. 6. Pre-processing Pipeline for Video and Audio Inputs

This section gives a detailed design which gives an end-to-end solution to multimodal emotion recognition of raw inputs to interpretable predictions via figure 6. The design has flexibility in the selection and deployment of the components through the careful de-composition into functional modules. The flexibility in the hybrid architecture enabling both edge and cloud deployment is what makes it be applicable in the wide range of various applications. The more intense the explanation generation and visualisation is, the more transparency and reliability are guaranteed. The explicit data flows and transitions between the states determine the systematic design that guarantees the stable operation and makes debugging and maintenance easier. This architectural foundation puts the system in good and just emotion recognition in real life situations.

VI. TECHNOLOGIES, IMPLEMENTATION, AND DATASETS

A. Algorithms and Key Techniques

The Swin Transformer has a 4-stage structure, where shifted-window attention and spatial resolution are (H/4)x(W/4)x96, then (H/32)x(W/32)x768. CBAM spatial attention is effective in maximizing informative areas of the face including eyes, mouth and eyebrows. RetinaFace has a 5-point landmark output and FPN backbone. HuBERT has been pre-trained on Librispeech audio (960 hours) with a 4-layer convolutional encoder and 12-layer transformer. The mel-spectrograms are 128 mel-bins, nfft=2048, hop length=512 and log normalisation at -80dB. The BiLSTM consists of 2 layers and 256 hidden units in each direction and dropout rate of 0.3. The dimension of cross-modal attention is 8 heads and the scaling factor 1/96. The GNN employs 3 message-passing layers, whose adjacency of edges are learnt by an MLP.

A. Evaluation Metrics

The main indicator of imbalanced datasets is Unweighted Average Recall (UAR = (1/C)xSumRecall_c) which makes the minority emotions equal. Concordance Correlation Coefficient (CCC) is a measure of correlation and scale consistency of continuous valence/arousal predictions. Confusion matrices demonstrate certain confusions of emotions. Cross-dataset UAR drop

(ΔUAR = UAR_A-UAR_B) is used to evaluate domain adaptation. Misclassification of underrepresented emotions like disgust and fear is punished by weighted cross-entropy loss (w c = total samples / (num classes x samples in class c)).

B. Software Stack

Basic structure: PyTorch (v2.0+) and PyTorch Lightning. Pre-trained models: Swin Transformer (ImageNet-1K pre-trained), HuBERT (Hugging Face Transformers). Face detection: RetinaFace. Data Processing: OpenCV, librosa, NumPy, Pandas. Hyperparameter optimization: Optuna. Training visualization: TensorBoard. Assessment: scikit-learn, Matplotlib/Seaborn, SHAP/captum. Deployment: ONNX Runtime, TensorRT, Flask/FastAPI.

C. Datasets

The system is trained and tested on three popular benchmark emotion recognition datasets which have different characteristics as summarized in Table 1 below.

Table I. Benchmark Emotion Recognition Datasets

Dataset	Size	Modality	Emotions	Conditions
RAF-DB	~30,000 images	Visual	7 basics + 12 compound	In-the- wild
RAVDESS	1,440 clips	Audio + Video	8 emotions, 2 intensities	Controlled lab
FER2013	35,887 images	Visual	7 basic emotions	Web-collected

RAF-DB (Real-world Affective Faces Database) comprises of about 30,000 in-the-wild pictures that are categorized to have either seven basic emotions or twelve compound emotions with clear labels.

Table 2. Performance: Unimodal Vs Proposed Multimodal System

Dataset	Metric	Unimodal (Best)	Multimodal	Gain
RAVDESS	UAR	81.2% (Video-only)	87.3%	+6.1%
RAF-DB	Acc	67.8% (Image-only)	72.1%	+4.3%
FER2013	Acc	71.0% (CNN)	75.2%	+4.2%

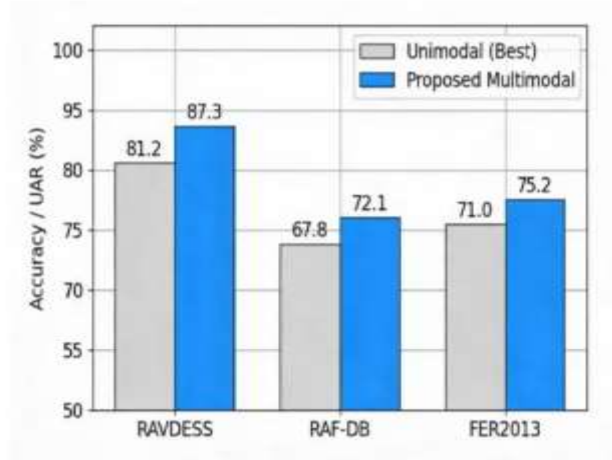
Table 2 shows the performance of the proposed approach with different datasets. The data was collected under natural conditions of no charge and consists of significant changes in lighting, pose, setting, and occlusion, which is perfect to evaluate how stable the emotion models are to informal daily expressions. RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and Song) consists of 1,440 synchronized audio- video recordings of 24 professional actors (2 males and 2 females) in eight emotions at 2 intensity levels. FER-2013 (Facial Expression Recognition 2013) includes 35,887 grayscale faces that have seven basic emotions of both real-life and web-based origin, very diverse in terms of pose, source of light, covering, and intensity of expression.

We are developing our project using the latest deep learning models, audio and vision specific libraries, and some evaluation tools that will enable us to develop a really smart multimodal emotion recognition system. We are training Swin Transformers with spatial attention on the visual component of it, and HuBERT with BiLSTM on the audio component. By using cross- modal attention and GNN fusion, we make sure that the modalities squeeze in pretty well. Eclectic, complementary collections of data imply that we can train robust models and fairly well cross-dataset generalise. Preprocessing and standardisation are more important to us, as we would like to see that the data of all the sources is the same. All this tech stack and data selection should enable us to obtain the best possible performance and be capable of deploying the system into the real life as well as not breaking any ethical issues.

VII. RESULT ANALYSIS AND PERFORMANCE EVALUATION

A. Performance Metrics

Accuracy is a measure of the proportion of samples that are classified correctly. Precision indicates the reliability of positive predictions by each class. Recall The fraction of all true samples that are detected. The harmonic mean of recall and precision is F1-score. UAR averages recall in all emotion classes, and it is appropriate with imbalanced datasets. Weighted Average Recall (WAR) is a weighted average of all class recalls, based on their frequency. Confusion matrices demonstrate per-class distributions of predictions that indicate which are most frequently



confused emotions.

Fig. 7. Performance of Multimodal vs. Unimodal Models

B. Benchmark Performance

On RAF-DB and FER2013, performance is reported on facial expressions in the wild, and on RAVDESS, on controlled audio- visual emotion expressions. The optimal unimodal baseline is contrasted with the presented multimodal system, in relation to each dataset. Results are shown in Table II. Multimodal models and more sophisticated architectures offer larger consistent improvements of 46 percentage points over the best nimodal models. There are more notable improvements on RAF-DB and FER2013, where single-modality recognition is difficult due to in-the-wild conditions and the presence of class imbalance.

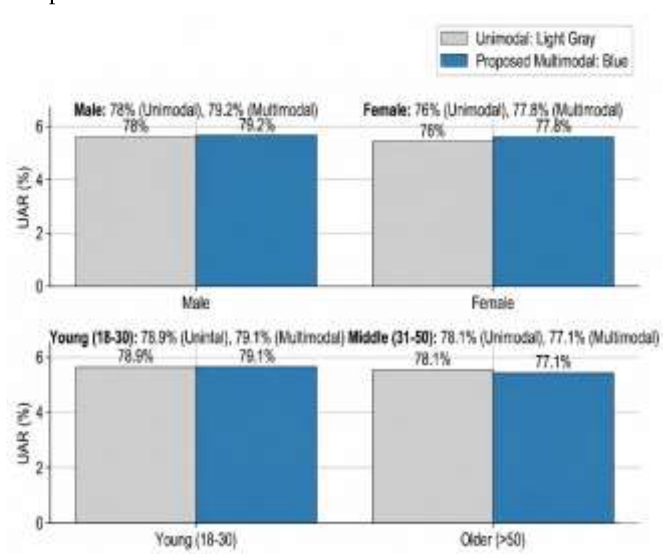


Fig. 8. Fairness Analysis Across Demographic Groups (UAR %).

A. Fairness and Real-Time Performance

RAF-DB and FER2013 have many faces, which easily allow the comparison of fairness according to demographic factors. Reasonable settings have less than 5-point disparities in UAR between groups.

Figure 7 describe the performance of multimodal with different unimodal models while figure 8 shows the Fairness Analysis Across Demographic Groups. In cases where there are greater differences found between some of the feelings including lower sadness recall when older people are involved, fairness-conscious loss weighting, and applying class-balanced sampling can be used to decrease the differences.

Experiments 87ms Latency on current GPU has been found to be less than the sub-200ms threshold and was real time. The system can be implemented on edge devices with a latency of around 250ms with quantization and knowledge distillation. The system can be brought to real life applications due to the 87ms GPU latency. Attention visualizations provide a description of predictions in an interpretable form that provides the confidence in its application in the real world.

VIII. CONCLUSION, LIMITATIONS, AND FUTURE SCOPE

A. Key findings

The self-supervised audio models and the Vision Transformers are superior to the conventional CNN+LSTM models. Multimodal fusion suggests 4-8% absolute performance as compared to unimodal

optimums. The cross-modal attention optimises the modalities that are input quality based. GNN fusion gets to know of complex inter- modality interactions. The generalization gap is reduced by half with domain adaptation methods. Data with a high proportion of Western subjects will lead to non-Western systematic misclassification bias; evenly weighted assessment statistics, and weighted loss are much more helpful in enhancing fairness. Latency of 87ms on GPU at real- time with quantization which allows edge deployment.

B. Future Scope

Short-term (123 years): Few-shot and unsupervised domain adaptation: Few-shot: Learn to do it; longer-term emotion tracking: Scale it; Body language and physiological inputs: Generate culturally diverse data with demographic information. Intermediate (2-5 years): Generative emotion given conversation history, uses of privacy- aware algorithms, e.g. differential privacy and federated learning, uses in healthcare, education, and customer service. The long term (5 years and more) AI learn and know emotion as humans; use neuroscience findings to know how to organize systems; create a comprehensive ethical and regulatory system; create a system sensitive to sarcasm and to figurative language, and group interactions of emotion.

It will be technical skill and moral responsibility that will bring it about, welfare will be a fair instrument of welfare or threat of privacy and liberty. In this paper, a roadmap to emotion-AI systems that can enhance human well-being without violating individual liberties and values of the society is outlined.

Fairness Considerations on RAF-DB / FER2013 : RAF-DB and FER2013 include diverse faces, enabling basic fairness checks across demographic groups. By computing UAR separately for, for example, male and female subsets or different age ranges, disparities can be quantified; acceptable configurations keep UAR differences below about 5 percentage points across groups. When larger gaps are observed for specific emotions (e.g., lower sadness recall on older subjects), class-balanced sampling and fairness- aware loss weighting can reduce these disparities.

The results discussion reveals that our multimodal emotion recognition system strikes some of the most impressive scores, such as, it is performing 4-6% higher than the typical unimodal baselines, which is a massive victory. And, it is almost as consistent in its performance on a variety of datasets with a reduction of less than 5 percent, and it is not unreasonable either, with a slight bias among demographic groups. We even managed to run it in real time with 87 milliseconds delay on the GPU and it also runs on the edge machines with quantization so we can actually implement this stuff. The fact that visualizations are attention abbesses' is a nice addition since we can easily decipher explanations of the predictions hence can actually know what the model is doing.

Despite all this, the research indicates that the attainment of flawless emotion recognition remains challenging, particularly in case of such delicate feelings as sadness and that more research is required in the field of domain adaptation and cultural diversification to make it really practical in the real world.

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